COMPUTER ORIENTED STATISTICS

Unit-II: Probability

Comprehensive Study Notes
HPU BCA Semester V

2.1 Introduction to Probability

Theory

Probability is a measure of the likelihood that an event will occur. It quantifies uncertainty and is expressed as a number between 0 and 1.

Key Definitions

Term	Definition	Example
Experiment	Any process that produces outcomes	Tossing a coin
Sample Space (S)	Set of all possible outcomes	{H, T}
Event (E)	Subset of sample space	Getting Head
Probability	P(E) = n(E)/n(S)	P(H) = 1/2

Types of Probability

Type	Description	Formula
Classical (A Priori)	Based on logical analysis, equally likely outcomes	P(E) = n(E)/n(S)
Empirical (A Posteriori)	Based on actual experiments and observations	P(E) = Number of times E occurred / Total trials
Axiomatic	Based on set of axioms defined by Kolmogorov	Follows axioms of probability

Classical Probability

$$P(E) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}} = \frac{n(E)}{n(S)}$$

Characteristics:

- All outcomes are **equally likely**
- Based on **logical analysis**
- Also called **A Priori Probability**

Example: Probability of getting an even number when rolling a die

```
S = {1, 2, 3, 4, 5, 6}, n(S) = 6
E = {2, 4, 6}, n(E) = 3
P(E) = 3/6 = 1/2
```

Empirical Probability

$$P(E) = \frac{\text{Number of times E occurred}}{\text{Total number of trials}}$$

Characteristics:

- Based on **actual experiments**

- Also called **A Posteriori Probability**
- More trials → More accurate

Example: If a coin is tossed 100 times and heads appear 48 times:

```
P(Heads) = 48/100 = 0.48
```

Axioms of Probability (Kolmogorov's Axioms)

Axiom	Statement
Axiom 1 (Non-negativity)	$P(E) \geq 0$ for any event E
Axiom 2 (Normalization)	$P(S) = 1$ for the sample space S
Axiom 3 (Additivity)	For mutually exclusive events $E_1, E_2, \ldots$: $P(\cup E_i) = \sum P(E_i)$

2.2 Addition Theorem of Probability

Theory

The Addition Theorem helps find the probability of the **union** of two or more events.

For Mutually Exclusive Events

If events A and B are mutually exclusive (cannot occur simultaneously):

$$P(A \cup B) = P(A) + P(B)$$

Example: Drawing a card from a deck

```
Event A = Drawing a heart
Event B = Drawing a diamond
Since A and B are mutually exclusive:
P(A u B) = P(A) + P(B) = 13/52 + 13/52 = 26/52 = 1/2
```

For Non-Mutually Exclusive Events

If events A and B can occur simultaneously:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Example: Drawing a card from a deck

```
Event A = Drawing a Heart (13/52)
Event B = Drawing a King (4/52)
A n B = Heart King (1/52)

P(A u B) = 13/52 + 4/52 - 1/52 = 16/52 = 4/13
```

For Three Events

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

Complete Example

Example: A card is drawn from a well-shuffled deck of 52 cards. Find probability of getting a red card or a king.

```
Let A = Event of getting a red card (26 cards)
Let B = Event of getting a king (4 cards)
A n B = Event of getting a red king (2 cards)

P(A u B) = P(A) + P(B) - P(A n B)
         = 26/52 + 4/52 - 2/52
         = 28/52
         = 7/13
```

2.3 Multiplication Theorem of Probability

Theory

The Multiplication Theorem helps find the probability of the **intersection** of two or more events.

For Independent Events

If events A and B are independent (occurrence of one doesn't affect the other):

$$P(A \cap B) = P(A) \times P(B)$$

Example: Tossing two coins

```
A = Getting head on first coin = 1/2
B = Getting head on second coin = 1/2
P(A n B) = P(HH) = 1/2 x 1/2 = 1/4
```

For Dependent Events

If events A and B are dependent:

$$P(A \cap B) = P(A) \times P(B|A)$$

Where P(B|A) is the **conditional probability** of B given A has occurred.
Example: Drawing two cards without replacement

```
A = First card is a king = 4/52
B|A = Second card is a king (given first was king) = 3/51
P(A n B) = 4/52 x 3/51 = 12/2652 = 1/221
```

Generalization for Multiple Events

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) = P(A_1) \times P(A_2|A_1) \times P(A_3|A_1 \cap A_2) \times \ldots$$

2.4 Conditional Probability

Theory

Conditional Probability is the probability of an event occurring given that another event has already occurred.

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{where } P(B) > 0$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \quad \text{where } P(A) > 0$$

Important Results

1. $P(A|B) = \frac{P(A \cap B)}{P(B)}$
2. $P(B|A) = \frac{P(A \cap B)}{P(A)}$
3. $P(A \cap B) = P(A) \times P(B|A) = P(B) \times P(A|B)$

Example: Conditional Probability

Problem: In a class of 100 students: 60 like Mathematics, 40 like Physics, 30 like both. Find: (i) P(Physics | Mathematics) (ii) P(Mathematics | Physics)

```
Let M = Event of liking Mathematics, P(M) = 60/100 = 0.6
Let P = Event of liking Physics, P(P) = 40/100 = 0.4
P(M n P) = 30/100 = 0.3

(i) P(P|M) = P(P n M)/P(M) = 0.3/0.6 = 0.5
(ii) P(M|P) = P(M n P)/P(P) = 0.3/0.4 = 0.75
```

Conditional Probability Table

	Likes Math	Likes Physics	Total
Loves Both	-	-	30
Loves Math Only	-	-	30
Loves Physics Only	-	-	10
Neither	-	-	30
Total	60	40	100

2.5 Independent Events

Theory

Two events A and B are **independent** if the occurrence of one does not affect the probability of occurrence of the other.

Conditions for Independence

Condition	Formula
Definition	$P(A \cap B) = P(A) \times P(B)$
Conditional	$P(A\|B) = P(A)$ and $P(B\|A) = P(B)$

Example: Checking Independence

Problem: A die is rolled twice. Let A be event of getting 1 on first roll, B be event of getting 2 on second roll.

```
P(A) = 1/6, P(B) = 1/6, P(A n B) = 1/36
P(A) x P(B) = 1/6 x 1/6 = 1/36 = P(A n B)
Therefore, events are independent.
```

Properties of Independent Events

Property	Statement
1	If A and B are independent, A and $\bar{B}$ are independent
2	If A and B are independent, $\bar{A}$ and B are independent
3	If A and B are independent, $\bar{A}$ and $\bar{B}$ are independent

Pairwise vs Mutual Independence

Type	Description
Pairwise Independence	Events are independent when taken two at a time
Mutual Independence	All events are independent together (stronger condition)

2.6 Bayes' Theorem

Theory

Bayes' Theorem describes the probability of an event based on prior knowledge of conditions that might be related to the event. It helps in finding the **reverse probability**.

$$P(A_i|B) = \frac{P(A_i) \times P(B|A_i)}{\sum_{j=1}^{n} P(A_j) \times P(B|A_j)}$$

Law of Total Probability

$$P(B) = \sum_{j=1}^{n} P(A_j) \times P(B|A_j)$$

Example: Bayes' Theorem

Problem: In a factory, machines A, B, and C produce 25%, 35%, and 40% of items respectively. Their defective rates are 5%, 4%, and 2% respectively. If a randomly selected item is defective, what is the probability it was produced by machine A?

```
Let A = Item from Machine A, P(A) = 0.25
Let B = Item from Machine B, P(B) = 0.35
Let C = Item from Machine C, P(C) = 0.40
Let D = Item is defective
P(D|A) = 0.05, P(D|B) = 0.04, P(D|C) = 0.02

By Bayes' Theorem:
P(A|D) = P(A) x P(D|A) / [P(A)P(D|A) + P(B)P(D|B) + P(C)P(D|C)]
       = (0.25 x 0.05) / (0.25 x 0.05 + 0.35 x 0.04 + 0.40 x 0.02)
       = 0.0125 / (0.0125 + 0.0140 + 0.0080)
       = 0.0125 / 0.0345
       = 0.3623
```

Answer: 36.23% probability it came from Machine A.

2.7 Random Variables

Theory

A **Random Variable** is a variable whose possible values are numerical outcomes of a random phenomenon.

Types of Random Variables

Type	Description	Example
Discrete Random Variable	Takes countable number of distinct values	Number of heads in 3 coin tosses: {0, 1, 2, 3}
Continuous Random Variable	Takes infinite number of values in an interval	Height of students, temperature

Probability Distribution

A **Probability Distribution** is a function that gives the probabilities of occurrence of different possible outcomes for a random variable.

Discrete Probability Distribution

Example: Number of heads in 3 coin tosses

X (Heads)	0	1	2	3
P(X)	1/8	3/8	3/8	1/8

```
Properties:
1. 0 <= P(X) <= 1 for all x
2. Sum of all P(X) = 1
```

Continuous Probability Distribution

For continuous random variables, probability is described by a **probability density function (PDF)**.

$$P(a \leq X \leq b) = \int_a^b f(x)\,dx$$

Where $f(x)$ is the probability density function.

2.8 Complete Solved Examples

Example 1: Addition Theorem

Problem: In a college, 60% students study Mathematics, 50% study Physics, and 30% study both. What is the probability that a randomly selected student studies Mathematics or Physics?

```
Let M = Studies Mathematics, P(M) = 0.6
Let P = Studies Physics, P(P) = 0.5
P(M n P) = 0.3

P(M u P) = P(M) + P(P) - P(M n P)
         = 0.6 + 0.5 - 0.3
         = 0.8
```

Answer: 80% students study Mathematics or Physics.

Example 2: Multiplication Theorem

Problem: A bag contains 5 red and 3 blue balls. Two balls are drawn without replacement. Find probability that: (i) Both are red (ii) First is red, second is blue

```
(i) Both are red:
A = First ball is red = 5/8
B|A = Second ball is red (given first was red) = 4/7
P(Both red) = 5/8 x 4/7 = 20/56 = 5/14

(ii) First red, second blue:
A = First ball is red = 5/8
B|A = Second ball is blue (given first was red) = 3/7
P(Red then Blue) = 5/8 x 3/7 = 15/56
```

Example 3: Conditional Probability

Problem: In a class of 100 students, 60 like Math, 40 like Physics, 30 like both. Find P(Physics | Math).

```
P(P|M) = P(P n M)/P(M) = 0.3/0.6 = 0.5
```

Example 4: Independent Events

Problem: A and B are independent events with P(A) = 0.3 and P(B) = 0.4. Find P(A n B), P(A|B), P(B|A).

```
(i) P(A n B) = P(A) x P(B) = 0.3 x 0.4 = 0.12
(ii) P(A|B) = P(A) = 0.3 (since independent)
(iii) P(B|A) = P(B) = 0.4 (since independent)
```

Example 5: Multi-Step Probability

Problem: A box contains 4 red, 3 blue, and 5 green balls. Three balls are drawn without replacement. Find probability that: (i) All are same color (ii) All are different colors (iii) At least 2 are red.

```
Total balls = 4 + 3 + 5 = 12

(i) All same color:
P(All red)  = 4/12 x 3/11 x 2/10 = 24/1320
P(All blue) = 3/12 x 2/11 x 1/10 = 6/1320
P(All green)= 5/12 x 4/11 x 3/10 = 60/1320
P(All same) = (24 + 6 + 60)/1320 = 90/1320 = 3/44

(ii) All different colors:
P(RBG) = 4/12 x 3/11 x 5/10 = 60/1320
Number of arrangements = 3! = 6
P(All different) = 6 x 60/1320 = 360/1320 = 3/11

(iii) At least 2 red:
P(2 red, 1 other) = C(4,2) x C(8,1) / C(12,3) = 6 x 8 / 220 = 48/220
P(3 red) = C(4,3) / C(12,3) = 4/220
P(At least 2 red) = 48/220 + 4/220 = 52/220 = 13/55
```

Example 6: Quick Examples Bank

Coin Toss: Toss a fair coin 3 times.

```
S = {HHH, HHT, HTH, HTT, THH, THT, TTH, TTT}, n(S) = 8

Exactly 2 heads: {HHT, HTH, THH}, n = 3, P = 3/8
At least 1 head: All except TTT, n = 7, P = 7/8
```

Dice Roll: Roll two dice.

```
Sum = 7: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1), n = 6, P = 6/36 = 1/6
Same number: (1,1), (2,2), (3,3), (4,4), (5,5), (6,6), n = 6, P = 6/36 = 1/6
```

Drawing Cards: From a deck of 52 cards.

```
P(Spade) = 13/52 = 1/4
P(Face card) = 12/52 = 3/13
P(King or Heart) = 4/52 + 13/52 - 1/52 = 16/52 = 4/13
```

2.9 Key Formulas Summary

Probability Rules

Rule	Formula	When to Use
Classical	P(E) = n(E)/n(S)	Equally likely outcomes
Addition (Mutually Exclusive)	P(A u B) = P(A) + P(B)	Events can't occur together
Addition (General)	P(A u B) = P(A) + P(B) - P(A n B)	Events can occur together
Multiplication (Independent)	P(A n B) = P(A) x P(B)	Events don't affect each other
Multiplication (Dependent)	P(A n B) = P(A) x P(B|A)	Events affect each other
Conditional	P(A|B) = P(A n B)/P(B)	Event given condition

Important Theorems

Theorem	Formula
Bayes' Theorem	$P(A_i\|B) = \frac{P(A_i)P(B\|A_i)}{\sum P(A_j)P(B\|A_j)}$
Total Probability	$P(B) = \sum P(A_i)P(B\|A_i)$

Independent Events Properties

Property	Formula
Definition	P(A n B) = P(A) x P(B)
Conditional	P(A|B) = P(A), P(B|A) = P(B)
Complement	P(A n ¬B) = P(A) x P(¬B)
Union	P(A u B) = P(A) + P(B) - P(A)P(B)

Quick Reference Card

Term	Meaning
Sample Space	All possible outcomes
Event	Subset of sample space
Probability	$0 \leq P(E) \leq 1$
Mutually Exclusive	Cannot occur together
Independent	One doesn't affect the other
Conditional	Probability given condition

Previous Year Questions

Short Questions (2 marks)

Q1. Define probability and its properties.

Q2. What is sample space?

Q3. Define mutually exclusive events.

Q4. What is conditional probability?

Q5. Define independent events.

Q6. State Addition Theorem of Probability.

Q7. State Multiplication Theorem of Probability.

Q8. What is the difference between dependent and independent events?

Q9. Define random variable. Give an example.

Q10 What is a probability distribution?

Medium Questions (5 marks)

Q1. Explain Addition Theorem with examples.

Q2. Explain Multiplication Theorem with examples.

Q3. What is conditional probability? Explain with an example.

Q4. Two dice are thrown. Find probability of getting sum 7 or sum 11.

Q5. A card is drawn from a deck. Find probability of getting a king or a heart.

Q6. Explain Bayes' Theorem with an example.

Q7. A bag contains 4 red and 6 black balls. Two balls are drawn without replacement. Find probability that both are red.

Long Questions (12 marks)

Q1. (a) State and prove Addition Theorem of Probability.
(b) A card is drawn from a well-shuffled deck. Find probability of getting a red card or a queen.

Q2. (a) Define independent events. Prove that if A and B are independent, then A and ¬B are also independent.
(b) In a factory, three machines produce items with given defective rates. Find probability using Bayes' Theorem.

Q3. (a) State Bayes' Theorem and explain with an example.
(b) Define random variables and explain discrete and continuous probability distributions.

Q4. (a) State and prove Multiplication Theorem of Probability for dependent and independent events.
(b) A box contains balls of different colors. Find probabilities of various combinations when drawing without replacement.